

Practice exam 2

Units 3 & 4 Paper 2

Perusal time: 5 minutes

Working time: 90 minutes

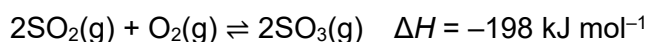
Section 1: Short response

Instructions

- If you need more space for a response, use the additional pages at the back of the book.
 - On the additional pages, write the question number you are responding to.
 - Cancel any incorrect response by ruling a single diagonal line through your work.
 - Write the page number of your alternative/additional response, i.e. “See page ...”
 - If you do not do this, your original response will be marked.
 - This paper has **5 questions** and is worth **37 marks**.
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QUESTION 1 (9 marks)

Consider the following reversible reaction for the key step in the contact process:



- a** Describe the key characteristic of the macroscopic properties of a system at chemical equilibrium, and contrast this with its microscopic state. *[2 marks]*

- b** Apply your understanding of Le Châtelier's principle to propose the levels of temperature and pressure (high or low) needed to maximise the yield of sulfur trioxide (SO_3) when equilibrium is established. Provide justification for each proposal. [3 marks]

- c** In an experiment, 4.00 mol of SO_2 and 4.00 mol of O_2 were mixed and allowed to come to equilibrium in a sealed 10.0 L vessel at a constant temperature. Once equilibrium was established, it was found that the vessel contained 2.00 mol of SO_3 .

- i** Determine the equilibrium constant expression (K_c) for the reaction. [1 mark]

- ii** Calculate the value of the equilibrium constant (K_c) at this temperature. [3 marks]

QUESTION 2 (5 marks)

Two straight-chain liquid fuels, pentan-1-ol and pentane, are investigated for their potential as energy sources.

- a** Determine the balanced chemical equation for the complete combustion of pentane (C_5H_{12}). [1 mark]

- b** The molar heat of combustion (ΔH°_c) for pentan-1-ol is $-3331 \text{ kJ mol}^{-1}$, and for pentane is $-3536 \text{ kJ mol}^{-1}$. Explain why pentan-1-ol has a less negative (lower magnitude) molar heat of combustion than pentane. [4 marks]

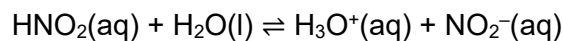
QUESTION 3 (3 marks)

A laboratory experiment involves separating a mixture of two organic compounds, butan-1-ol (C_4H_9OH) and butanal (C_3H_7CHO), by column chromatography with a silica gel column (stationary phase: polar) and a hexane/ethyl acetate solvent mixture (mobile phase: relatively non-polar).

Predict which compound, butan-1-ol or butanal, will elute first from the chromatography column. Justify your answer in terms of intermolecular forces and separation theory.

QUESTION 4 (8 marks)

Nitrous acid (HNO_2) is a weak acid used in chemical synthesis. Its acid ionisation constant, K_a , is 4.0×10^{-4} at 25°C .



- a** Determine the equilibrium constant expression (K_a) for the ionisation of nitrous acid. [1 mark]

- b** Calculate the pH of a 0.50 M solution of nitrous acid at 25°C . [4 marks]

- c** Calculate the percentage ionisation of the 0.50 M nitrous acid solution. [3 marks]

QUESTION 5 (12 marks)

An unknown straight-chain organic compound, Compound X, has the empirical formula $C_3H_6O_2$. Analysis suggests it is an ester.

- a** Calculate the molecular formula of Compound X if its molar mass is 74.09 g mol^{-1} . [3 marks]

- b** The ^1H NMR spectrum of Compound X shows three distinct proton environments (chemical shift) with area ratios of 3:2:1. The infrared spectrum shows a strong absorption peak at $1,730 \text{ cm}^{-1}$ and no broad absorption between $3,200$ and $3,600 \text{ cm}^{-1}$. Draw the condensed structural formula for Compound X and state its IUPAC name. [5 marks]
